

Volume with Whole Number Edges

Lesson 10-1

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

A box $3 \times 2 \times 4$ holds 24 unit cubes, so $V = 24$ cubic units

Volume

How much space is inside a solid shape.

A cereal box or shoe box — it has length, width, and height

Rectangular prism

A solid box shape with six flat rectangle sides.

A tiny cube that is $1 \text{ in} \times 1 \text{ in} \times 1 \text{ in} = 1 \text{ in}^3$ (one cubic inch)

Cubic units

The units used to measure space inside, like cubic inches.

$V = l \times w \times h$: for a box $5 \times 3 \times 2$, volume = 30 cubic units

Length, width, height

How long, how wide, and how tall a box is.

A cross-shaped pattern of 6 rectangles folds into a rectangular box

Net

A flat shape that folds up into a solid.

A cube has 12 edges — 4 along the top, 4 along the bottom, 4 vertical

Edge

The line where two flat sides of a solid meet.

Key Ideas & Notes

- Your team is building time capsule containers to store class memories.
- Each container is a rectangular prism, and you need to figure out how much each one can hold.
- That means calculating the volume using whole number dimensions!
- Calculate the volume of each time capsule container. Use $V = l \times w \times h$.

Think About It

- What three measurements do you need to find the volume of a box?
- What does volume tell us about a container?
- How is volume different from the area of one face?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Your time capsule container is a rectangular prism. To find how much it can hold, why do you need three measurements (length, width, and height) instead of just one or two?

Level 1 support

Start here: Volume measures space in three directions, so you need length, width, and height.

I need three measurements because volume fills space in ____ directions.

Necesito tres medidas porque el volumen llena el espacio en ____ direcciones.

Length and width alone only give the ____, not the volume.

El largo y el ancho solos solo dan el ____, no el volumen.

Word bank: volume length width height rectangular prism

Listen for: Students explain that volume is three-dimensional, so all three edges (length, width, height) are needed; two measurements only give area of one face.

Level 2

Could two time capsules have the same volume but different shapes? Justify with an example.

- Two capsules ____ have the same volume but different shapes because ____.
- For example, $___ \times ___ \times ___$ and $___ \times ___ \times ___$ both equal ____.

EXPLORE

Container C is $8 \times 3 \times 4$ and Container B is $5 \times 5 \times 2$. How did you count or multiply the unit cubes to find each volume, and which holds more?

Level 1 support

Start here: Multiply length $\times$ width $\times$ height to count all the unit cubes that fill the box.

I found the volume by multiplying $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$, **which equals** $\underline{\hspace{1cm}}$ **cubic units.**

Hallé el volumen multiplicando $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$, que es $\underline{\hspace{1cm}}$ unidades cúbicas.

Container $\underline{\hspace{1cm}}$ **holds more because its volume is** $\underline{\hspace{1cm}}$.

El contenedor $\underline{\hspace{1cm}}$ contiene más porque su volumen es $\underline{\hspace{1cm}}$.

Word bank: volume cubic units length width height rectangular prism

Listen for: Students compute Container C = $8 \times 3 \times 4 = 96 \text{ in}^3$ and Container B = $5 \times 5 \times 2 = 50 \text{ in}^3$, and conclude C holds more by connecting volume to multiplying all three edges.

Level 2

Container B (50 in^3) has a larger length and width than Container D ($10 \times 2 \times 3 = 60 \text{ in}^3$), yet D holds more. Explain how that is possible.

- D holds more even though $\underline{\hspace{1cm}}$ because volume depends on $\underline{\hspace{1cm}}$.
- A bigger length or width does not always mean a bigger volume because $\underline{\hspace{1cm}}$.

CONNECT

A moving company's large box is $24 \times 18 \times 18$ in and its medium box is $18 \times 14 \times 12$ in. Talk through how you would find how many more cubic inches the large box holds.

Level 1 support

Start here: Find each volume with $V = l \times w \times h$, then subtract the smaller from the larger.

The large box volume is ____ in^3 because $24 \times 18 \times 18 =$ ____.

El volumen de la caja grande es ____ in^3 porque $24 \times 18 \times 18 =$ ____.

The large box holds ____ more cubic inches because ____ - ____ = ____.

La caja grande contiene ____ pulgadas cúbicas más porque ____ - ____ = ____.

Word bank: volume cubic units rectangular prism length width height

Listen for: Students compute large = 7776 in^3 , medium = 3024 in^3 , then subtract to get 4752 in^3 more, and reason that you must find each volume before comparing.

Level 2

Why can't you just compare one dimension (like length) to decide which box holds more? Use the two boxes to justify.

- Comparing only length is not enough because ____.
- Even though the medium box is shorter in one edge, ____.

REFLECT

A classmate found the volume of a box and wrote the answer as 220 in^2 . How would you correct the units, and how do you explain why volume uses cubic units?

Level 1 support

Start here: Volume measures three-dimensional space, so the units must be cubic (in^3), not square (in^2).

The units should be ____ instead of in^2 because volume measures ____.

Las unidades deben ser ____ en lugar de in^2 porque el volumen mide ____.

Square units (in^2) are used for ____, but cubic units (in^3) are used for ____.

Las unidades cuadradas (in^2) se usan para ____, pero las unidades cúbicas (in^3) se usan para ____.

Word bank: volume cubic units area rectangular prism edge

Listen for: Students change in^2 to in^3 and explain that volume fills 3D space (three edges multiplied), while square units describe flat area of one face.

Level 2

If you multiplied inches $\times$ inches $\times$ inches, why does that produce cubic inches?

Generalize the rule for any volume unit.

- Multiplying three lengths gives cubic units because ____.
- In general, the unit for volume is always ____ because ____.

Guided Examples

Example 1

What is the volume of a rectangular prism with $l = 7$ in, $w = 3$ in, $h = 4$ in?

Solution: $V = l \times w \times h = 7 \times 3 \times 4 = 84$ cubic inches.

Answer: A. 84 in^3

Example 2

A time capsule box has a volume of 120 cm^3 . Its length is 10 cm and width is 4 cm. What is its height?

Solution: $V = l \times w \times h \rightarrow 120 = 10 \times 4 \times h \rightarrow 120 = 40h \rightarrow h = 3 \text{ cm}$.

Answer: A. 3 cm

Example 3

Which unit is used for volume?

Solution: Volume measures 3D space, so it uses cubic units like in^3 . Square units (in^2) are for area, and inches are for length.

Answer: A. Cubic inches (in^3)

Write About the Math

The Writing Revolution

I can explain my work using the words volume, rectangular prism, cubic units, and edge.

1. Kernel Sentence subject + verb

Model: Volume is how much space is inside a solid shape.

Volumen es cuánto espacio hay dentro de una figura sólida.

Write a kernel sentence about volume. Use a subject and a verb.

Escribe una oración base sobre volumen. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Volume matters in math

Volumen importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Volume matters in math because ____.

Volumen importa en matemáticas porque ____.

but
pero

Volume matters in math, but ____.

Volumen importa en matemáticas, pero ____.

so
entonces

Volume matters in math, so ____.

Volumen importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about volume.

Di un hecho verdadero sobre volume.

Volume ____.

Question *Pregunta*

Ask a question about volume.

Haz una pregunta sobre volume.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about volume.

Muestra entusiasmo sobre volume.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with volume.

Dile a un compañero qué hacer con volume.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

I need three measurements because volume fills space in ____ directions.

Necesito tres medidas porque el volumen llena el espacio en ____ direcciones.

Length and width alone only give the ____, not the volume.

El largo y el ancho solos solo dan el ____, no el volumen.

I found the volume by multiplying ____ × ____ × ____, which equals ____ cubic units.

Hallé el volumen multiplicando ____ × ____ × ____, que es ____ unidades cúbicas.

Try It

Solve on your own. Check the answer key when you are done.

1. Which unit is used for volume?

- A. Cubic inches (in^3)
- B. Square inches (in^2)
- C. Inches (in)
- D. Degrees ($^\circ$)

Show your work:

2. Two storage bins: Bin A is $8 \times 6 \times 5$ inches. Bin B is $9 \times 4 \times 7$ inches. Which holds more and by how much?

- A. Bin B by 12 in^3
- B. Bin A by 12 in^3
- C. Bin B by 24 in^3
- D. They hold the same

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A box needs to hold exactly 60 cubic inches. Give three different sets of whole-number dimensions that work. Which set would make the box closest to a cube shape? Why might that matter?

Sentence starter: Option 1: $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$. Option 2: $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$. Option 3: $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$.

The $\underline{\hspace{1cm}}$ option is closest to a cube because $\underline{\hspace{1cm}}$. This matters because $\underline{\hspace{1cm}}$.

Show your work:

Reflect — Exit Ticket

A rectangular prism has $l = 11$ in, $w = 5$ in, $h = 4$ in. What is the volume?

- A. 220 in^3
- B. 55 in^3
- C. 220 in^2
- D. 200 in^3

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. Cubic inches (in^3) — *Volume measures 3D space, so it uses cubic units like in^3 . Square units (in^2) are for area, and inches are for length.*
2. **Try It 2:** A. Bin B by 12 in^3 — *Bin A: $8 \times 6 \times 5 = 240 \text{ in}^3$. Bin B: $9 \times 4 \times 7 = 252 \text{ in}^3$. Bin B holds 12 in^3 more.*
3. **Exit Ticket:** A. 220 in^3 — *$V = 11 \times 5 \times 4 = 220$ cubic inches. Remember: volume uses cubic units (in^3).*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Volume is how much space is inside a solid shape.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).